Pasindu Ranasinghe

Software Engineering Intern Candidate

Sri Lanka | +94 76 453 8580 | mamepasindu@gmail.com
linkedin.com/in/pasindu-ranasinghe-9a7315326 | github.com/Pasinduranasinghe2001

Professional Summary

Computer Engineering undergraduate at the University of Jaffna with hands-on experience developing full-stack web applications, desktop client-server systems, and real-time communication applications. Skilled in Java, Spring Boot, JavaScript, Node.js, PHP, Python, MySQL, MongoDB, REST APIs, WebRTC, Socket.IO, and Java Swing. Strong foundation in object-oriented programming, data structures and algorithms, database systems, computer networks, and software development. Experienced in collaborative development using Git and GitHub, with additional experience applying optimization techniques to software problems. Currently conducting undergraduate research on deep-learning-based archaeological looting detection using satellite imagery. Seeking a Software Engineering internship to contribute to real-world software development and strengthen engineering skills.

Education

B.Sc. (Hons) in Computer Engineering 2023 – Present
University of Jaffna, Faculty of Engineering *Expected Graduation: 2027*
Relevant Coursework: Software Engineering, Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Networks, Operating Systems, Artificial Intelligence, Machine Learning

G.C.E. Advanced Level – Physical Science 2021
R/ Ananda Maithreya Central College (AMCC), Balangoda
Physics – B | Chemistry – B | Combined Mathematics – B

Technical Skills

Languages: Java, JavaScript, Python, PHP, C, C++, SQL
Frameworks & Technologies: Spring Boot, Spring MVC, Node.js, React, HTML5, CSS3, REST APIs, WebRTC, Socket.IO, Java Swing, Tkinter
Databases: MySQL, MongoDB
Development Tools: Git, GitHub, Maven, Postman, IntelliJ IDEA, Visual Studio Code
Core Concepts: Object-Oriented Programming, Data Structures & Algorithms, Client-Server Architecture, RESTful API Development, CRUD Operations, Database Design, SDLC, Version Control, Socket Programming, Constraint Optimization

Projects

EQuipHub – Equipment Request Management System *Spring Boot, React, MongoDB, MySQL*

- Developed a full-stack equipment management platform supporting equipment requests, approval workflows, user management, and inventory-related operations.
- Implemented backend services and RESTful APIs using Spring Boot and integrated application data with MongoDB and MySQL.
- Collaborated within a five-member development team using Git and GitHub for source control, code integration, and team-based development.

WebRTC Video Conferencing Application *Node.js, WebRTC, Socket.IO, JavaScript*

- Developed a real-time video conferencing application supporting meeting creation, joining, waiting-room functionality, and multi-user video communication.
- Implemented peer-to-peer audio and video communication using WebRTC with real-time signaling through Socket.IO.
- Developed and deployed the Node.js signaling server and tested communication between multiple participants across different networks.

Hostel Management System *PHP, JavaScript, HTML, CSS, MySQL*

- Developed a web-based system for managing student information, hostel records, room allocation, and administrative operations.
- Implemented database-driven CRUD operations using PHP and MySQL and developed the user interface using HTML, CSS, and JavaScript.

ClassChat – Java Client-Server Chat Application *Java, Java Swing, Socket Programming*

- Developed a desktop client-server chat application supporting real-time message exchange through Java socket programming.
- Designed the graphical user interface using Java Swing and applied object-oriented programming principles to structure the application.

Designing an Intelligent Timetabling System *Python, Simulated Annealing, Tkinter*

- Developed an intelligent timetable optimization system that generates feasible schedules while satisfying lecturer, room, student-group, course-ordering, and availability constraints.
- Implemented a Simulated Annealing optimization algorithm with iterative search, temperature cooling, and probabilistic acceptance of candidate solutions.
- Developed a Tkinter GUI to run the optimization, display the generated timetable, and report hard- and soft-constraint violations.

Gold Bomb – Java Mini Game *Java 21, Java Swing, AWT, AudioSystem*

- Developed an interactive Java Swing game where players collect randomly positioned gold objects while avoiding bombs.
- Implemented event-driven gameplay, randomized object placement, dynamic score tracking, game-over handling, and restart/exit functionality.
- Used Swing timers for timed game events and integrated Java's AudioSystem and Clip APIs for background music and sound effects.

Research

Looting Detection Using Satellite Images *Undergraduate Research – Ongoing*

- Investigating deep-learning approaches for detecting archaeological site looting using satellite imagery.
- Studying the DAFA-LS dataset containing monthly satellite imagery collected across archaeological sites in Afghanistan.
- Addressing the limitation of approaches that depend on long-term satellite image time series by investigating detection using limited temporal observations.
- Developing experiments using different numbers of satellite-image frames ($K = 1, 2, 3, 5, 8$) and comparing single-frame, change-based, and short-temporal fusion approaches.
- Evaluating models using Precision, Recall, F1-score, ROC-AUC, IoU, and mAP, while investigating frame-selection and missing-frame scenarios.

Professional Memberships

- Student Member – Institute of Electrical and Electronics Engineers (IEEE)
- Student Member – Institution of Engineers, Sri Lanka (IESL)

References

Dr. U. Thanuja
Senior Lecturer – Grade II, Department of Computer Engineering
Faculty of Engineering, University of Jaffna
Email: thanuja@eng.jfn.ac.lk | Tel: +94 77 436 3167

Dr. (Mrs.) J. Jananie
Senior Lecturer – Grade II, Department of Computer Engineering
Faculty of Engineering, University of Jaffna
Email: jananie@eng.jfn.ac.lk | Tel: +94 21 206 0161